Say again we have

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

then a linear system of differential equations is something of the form:

$$\underbrace{x'(t)}_{n \times 1} = \underbrace{A(t)}_{n \times n} \underbrace{x(t)}_{n \times 1} + \underbrace{f(t)}_{n \times 1}$$

where $A(t)$ and $f(t)$ have any functions of t as entries.

When $f(t) = 0$, this system is <u>homogeneous</u>, and when $A(t)$ has only constants as entries (no t in any entry), we say the system has <u>constant coefficients</u>.

If $x(t)$ has only one function, i.e. this system was just a single equation, this would be same terminology for constant coefficients, homogeneous used there.

So this is just a generalization of those ideas.

We will mostly only solve systems which are homogeneous with constant coefficients. Might dabble slightly into nonhomogeneous, but that can be tough.

Had to use undetermined coefficients or variation of parameters with single equations to solve those which was often quite tough, long calculations if nothing else. System could be much worse!

We won't really try to solve nonconstant coefficient systems, we mostly avoided that with single equations so not really realistic to think we can solve systems which have those.

With both we might mention them from a speculative standpoint, where they might come up. But solving them not so much.

So our focus will be on systems of type:

$$x'(t) = Ax(t)$$

No $f(t)$ since homogeneous, and we write A instead of $A(t)$ to emphasize the fact that we have constant coefficients.

But how to solve? Well we suspected taking $x_j = c_j e^{kt}$ for each j (to be clear, same exponent,

i.e. same k for all, different constant multiples though perhaps). This makes sense because in $x'(t) = Ax(t)$ every entry on both sides will then be of type Ce^{kt} so realistic to think they could all be equal.

Still delicate since all n entries have to match which is a strong requirement but realistic to think for proper k and c_j's that it could happen.

Also indirectly shows why being homogeneous with constant coefficients is so important as if say $A(t)$ or $f(t)$ had a term like $\sin t$, this choice wouldn't work at all.

But for which k and which c_j's will this work? Remember all n derivatives need to match other side so this is quite delicate.

Say λ is a real eigenvalue of A with associated real eigenvector:

$$W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

Turns out then that

$$x(t) = Ce^{\lambda t} w = Ce^{\lambda t} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

is a solution.

I.e. $x_j(t) = Ce^{\lambda t} w_j$ for each j in $[1, n]$.

Note that since w is an eigenvector, any nonzero multiple of it is as well. So we don't really need the C as that could be absorbed by the w. But we do this on the thought that w is a specific eigenvector so we leave C in front to make sure it's generalized.

Why does this work? Note that

$$Ax(t) = A[Ce^{\lambda t} w] = Ce^{\lambda t}[Aw] = C\lambda e^{\lambda t} w$$

as $Aw = \lambda w$ due to w being eigenvector for A

On other side:

$$x(t) = Ce^{\lambda t} w \text{ ----> } x'(t) = C\lambda e^{\lambda t} w$$

just via normal differentiation as C and w don't depend on t.

So that shows it works.

Expanding, we showed an nth order differential equation can be viewed as a system of n first order linear differential equations. Going other way, thinking of this system as an nth order equation, we feel like it ought to have n linearly independent solutions, and so far this has shown only one, what about others?

Say the n eigenvalues of A are all real (not necessarily distinct): $\lambda_1, \lambda_2, \ldots, \lambda_n$ with n linearly independent eigenvectors: $v_1, v_2, \ldots, v_n$

By saying that, we are making the big assumption that A is diagonalizable over $\mathbb{R}$. Will deal with other cases later. In this case general solution is:

$$\sum_{j=1}^{n} c_j e^{\lambda_j t} v_j$$

We showed each individual term in summation worked so just like for single homogeneous equations will work for the sum.

The matrix whose columns are the $e^{\lambda_j t} v_j$ is called the <u>fundamental matrix</u> of the system.

Like we often do via matrices, writing the solutions as such is a good way to catalog all that information at once and studying that matrix is a good way to learn more about the system.

B9.5.1

$$x_1' = -4x_1 + 2x_2$$
$$x_2' = 2x_1 - x_2$$

So here:

$$A = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$$

---> $$A - \lambda I = \begin{bmatrix} -4-\lambda & 2 \\ 2 & -1-\lambda \end{bmatrix}$$

meaning that

$$\det(A - \lambda I) = (-4 - \lambda)(-1 - \lambda) - 4 \stackrel{?}{=} 0$$

----> $$\lambda^2 + 5\lambda = 0$$

----> $$\lambda = -5, 0$$

Checking both:

$$A + 5I = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

which gives an eigenvector of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

and then

$$A + 0I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \sim \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix}$$

and that gives an eigenvector of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Thus solution is:

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = c_1 e^{-5t} \begin{bmatrix} -2 \\ 1 \end{bmatrix} + c_2 e^{0t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

or written differently, general solution is:

$$x_1(t) = -2c_1 e^{-5t} + c_2$$

$$x_2(t) = c_1 e^{-5t} + 2c_2$$

This is how it always works when you have n linearly independent real eigenvectors. Whether A is 2 x 2 or 10 x 10, doesn't matter. Of course larger matrix means more calculation as always, but the fundamental idea does not change.

What if A is not diagonalizable over $\mathbb{R}$? Remember two things can cause that:

(1) A has repeated real eigenvalue(s) whose algebraic multiplicity do not match their geometric multiplicity.
(2) A has a nonreal eigenvalue

In the case of (1) when we had a single equation whose characteristic equation has a repeated real root r, our second solution after e^{rt} is of the type te^{rt}

Will work somewhat similarly for system, but unfortunately much harder to put that together. In short, you don't have the n linearly independent eigenvectors needed so those extra solutions come from the repeated eigenvalues, again things of type te^{rt}

But again how to design the new vector etc., much trickier. In HW Exercise B9.5.35, they outline how to do that for a 2 x 2 matrix with a repeated eigenvalue, but that's already tough to understand. Trying to generalize to larger matrices, just not accessible to us.

Especially the issue that if you have an eigenvalue whose algebraic multiplicity is say 5, its geometric multiplicity being 4 vs. 2 likely has to be handled quite differently, in the 2 x 2 case, hard as that is, it's always algebraic multiplicity 2 with geometric multiplicity 1 so at least consistent in that respect.

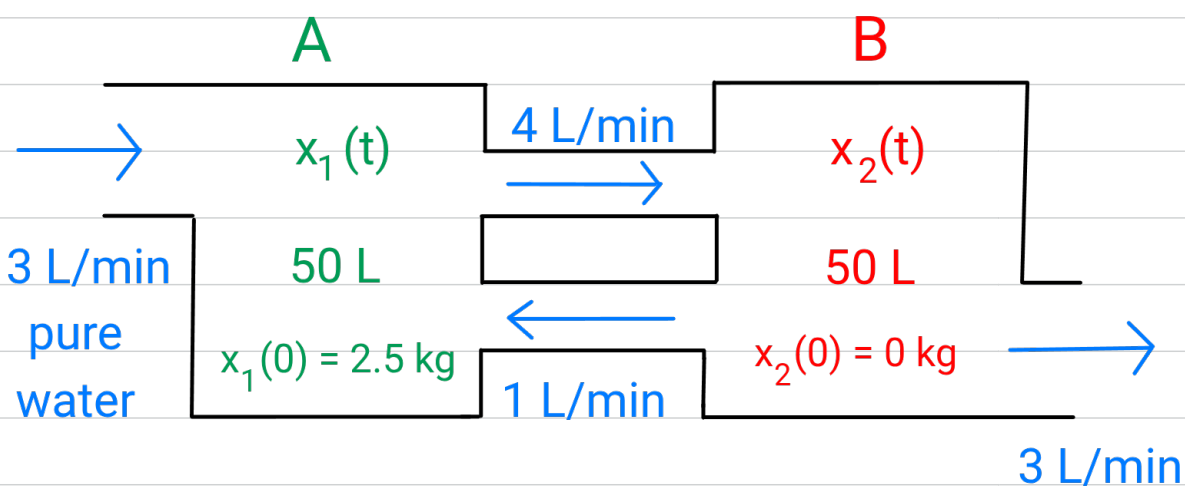
Again this is very hard to generalize so can't really do that in this class.

The other case, nonreal eigenvalues, we will handle in the next section.

That's actually easier to deal with, is like single equations whose characteristic equation has nonreal roots will use cos, sin terms.

Before next section, do one physical example of a mixing problem with two tanks which will be solved with the earlier mentioned ideas.

B9.5.45



$x_1(t)$ and $x_2(t)$ represent the amount of salt in in each respective tank at time t.

First note that the amount of liquid in each tank stays fixed at 50 L due to the inflow/outflow rates (that's very important!). But how to track the amount of salt

Just as when there was only one of tank, measure how much salt is entering tank at given moment minus how much is leaving. That is, rate in – rate out. Here that will give:

$$x_1' = \left(\frac{x_2(t)\ \text{kg}}{50\ \text{L}}\right)\left(\frac{1\ \text{L}}{\text{min}}\right) - \left(\frac{x_1(t)\ \text{kg}}{50\ \text{L}}\right)\left(\frac{4\ \text{L}}{\text{min}}\right)$$

$$x_2' = \left(\frac{x_1(t)\ \text{kg}}{50\ \text{L}}\right)\left(\frac{4\ \text{L}}{\text{min}}\right) - \left(\frac{x_2(t)\ \text{kg}}{50\ \text{L}}\right)\left(\frac{4\ \text{L}}{\text{min}}\right)$$

Now that units align, can write these more compactly as

$$x_1' = \frac{-4}{50} x_1 + \frac{1}{50} x_2 \quad , \quad x_1(0) = 2.5$$

$$x_2' = \frac{4}{50} x_1 - \frac{4}{50} x_2 \quad , \quad x_2(0) = 0$$

$$A = \begin{bmatrix} \frac{-4}{50} & \frac{1}{50} \\ \frac{4}{50} & \frac{-4}{50} \end{bmatrix}$$

Could factor out $\frac{1}{50}$ but that changes eigenvalues so generally alters nature of calculation so maybe do that while solving characteristic equation instead.

Then we get:

$$A - \lambda I = \begin{bmatrix} \frac{-4}{50} - \lambda & \frac{1}{50} \\ \frac{4}{50} & \frac{-4}{50} - \lambda \end{bmatrix}$$

And thus

$$\det(A - \lambda I) = \left(\frac{-4}{50} - \lambda\right)^2 - \frac{4}{50} \cdot \frac{1}{50} \stackrel{?}{=} 0$$

---> $$(-4 - 50\lambda)^2 - 4 = 0$$

---> $$-4 - 50\lambda = \pm 2$$

---> $$\lambda = \frac{-3}{25}, \frac{-1}{25}$$

So now find eigenvectors:

$$A + \frac{3}{25}I = \begin{bmatrix} \frac{1}{25} & \frac{1}{50} \\ \frac{4}{50} & \frac{1}{25} \end{bmatrix} \sim \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$$

yielding an eigenvector of $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

and for the other:

$$A + \frac{1}{25}I = \begin{bmatrix} \frac{-1}{25} & \frac{1}{50} \\ \frac{4}{50} & \frac{-1}{25} \end{bmatrix} \sim \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}$$

and that gives eigenvector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Thus we have

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = c_1 e^{\frac{-3}{25}t} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 e^{\frac{-1}{25}t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

i.e.

$$x_1(t) = c_1 e^{\frac{-3}{25}t} + c_2 e^{\frac{-1}{25}t}$$

$$x_2(t) = -2c_1 e^{\frac{-3}{25}t} + 2c_2 e^{\frac{-1}{25}t}$$

Plugging in initial conditions yields:

$$2.5 = x_1(0) = c_1 + c_2$$

$$0 = x_2(0) = -2c_1 + 2c_2 \text{ ----> } c_1 = c_2$$

$c_1 = c_2 = 1.25$

Thus final answer is:

$$x_1(t) = 1.25e^{\frac{-3}{25}t} + 1.25e^{\frac{-1}{25}t}$$

$$x_2(t) = -2.5e^{\frac{-3}{25}t} + 2.5e^{\frac{-1}{25}t}$$

What happens to these as t ----> ∞ ?

Well all exponential terms have negative exponents so those all go to 0. And that makes sense since tanks only being replenished with pure water, so concentration of salt ought to go to 0 in the longterm.

But realistically we would probably not want that to happen in a real world setting. Say tanks have fish in them, they wouldn't survive longterm then.

How to avoid that? Natural way is to replenish tank with water which has some concentration of salt in it.

If we do that, how does it change how this solves? Well for the rate in for one or both there will be a constant in addition to the term which represents the salt it receives from other tank.

In short, the system will become nonhomogeneous then, harder to solve.

Ok, maybe still doable since the $f(t)$ for the nonhomogeneous part is just a constant, not containing some more annoying term like $\sin t$. But still harder.

Also, what if amount of liquid entering one or both tanks doesn't match what's leaving, i.e. volume of liquid doesn't stay fixed? If say Tank A gains 1 L/min, then its volume at time t is $50 + t$, not the constant 50.

We would then have that $50 + t$ in denominator of some coefficients so this would no longer be constant coefficients, and thus not really accessible to us to solve probably.

We dealt with this situation when there was only one tank. We had to use Integrating Factor then, was harder as with constant volume we just used separability, which is usually simpler and easier to understand.

But ok, it was still solvable in that circumstance, even if it was harder.

But once we have this in a system, not really doable for us to solve it most likely.